TECHNISCHE UNIVERSITÄT DRESDEN

Fakultät Elektrotechnik und Informationstechnik
Institut für Nachrichtentechnik
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Task for Master Thesis of

Mr. Chongyang Zheng Academic year: ET/ , matriculation number:

Topic: Randomized Identification Codes with a Focus on K-Identification

Task description:

Identification via channels (ID) asks receivers to decide whether a specific message was sent, rather than to reconstruct it. Randomized ID codes achieve doubly-exponential message sets in blocklength and enable vanishing Type-I (false-accept) and Type-II (miss) error probabilities at positive ID rate. K-Identification (K-ID) generalizes this: the receiver asks whether the transmitted message belongs to a set of K candidates. K-ID is relevant for goal/semantics-driven communication (e.g., digital twins), where the action depends only on whether the current state falls into a small set of actionable hypotheses, not on exact message recovery. This thesis will study randomized ID codes with emphasis on K-ID, and extend the ideas and evaluation from “Stopping the Data Flood: Post-Shannon traffic reduction in digital-twins applications” to the K-ID setting. The work should balance theory, algorithms, and software-driven experiments using our provided library.

The student is expected to:

Survey the state of the art:

- Core ID concepts: randomized encoders/acceptance regions, ID capacity, error exponents, and operational interpretations; contrast with Shannon transmission.
- Define K-ID precisely (membership in a K-set) and position it vs. related notions (multiple-object ID, generalized ID; deterministic vs randomized variants.

Formulate the problem & metrics for K-ID:

- Formal model (channel class, randomization/common randomness budget, decoder tests).
- Report both Type-I/Type-II errors and computational metrics (encode/decode time, seed length, memory), plus finite-blocklength behavior.

Replicate & extend the digital-twins baseline to K-ID:

- Reproduce key experiments/results from the “Stopping the Data Flood” study (baseline ID) with our software stack.
- Design K-ID experiments that reflect digital-twin quantifying traffic reduction vs. reliability/latency.

Use our software library to build a modular K-ID test bench:

- Implement randomized ID/K-ID encoders and decoders (acceptance region tests) as pluggable components.

Design and run a simulation campaign:

- Measure KPIs such as: Type-I/II error curves, empirical exponents, queries per symbol, and compute/memory usage. Quantify traffic reduction vs. reliability/latency

Deliver a comparative assessment with clear takeaways:

- When do systems benefit from K-ID?
- Tradeoffs among randomization, complexity, and reliability.

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Responsible Tutor